

Research Paper Review Report

Paper Information

- Title: Improving Language Models by Retrieving from Trillions of Tokens
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Executive Summary

The paper introduces Retro, a retrieval-enhanced autoregressive language model that conditions on document chunks retrieved from a large corpus based on local similarity with preceding tokens. By leveraging a 2 trillion token database, Retro achieves comparable performance to GPT-3 and Jurassic-1 on the Pile, despite using 25× fewer parameters. This is made possible by a novel chunked cross-attention mechanism that efficiently integrates retrieved text into the model's predictions. The approach demonstrates the potential of augmenting language models with explicit memory at an unprecedented scale, offering a promising direction for future research in natural language processing.

The method's scalability is highlighted by its ability to provide constant gains for models ranging from 150M to 7B parameters. Moreover, the performance of Retro improves with the size of the retrieval database and the number of retrieved neighbors, showcasing the importance of the retrieval component. The paper also addresses the issue of test set leakage by proposing an evaluation methodology that considers the proximity of test documents to the training set, ensuring a more robust assessment of the model's capabilities.

Detailed Analysis

Problem Statement

Traditional language models rely on increasing model size and training data to improve performance, which leads to higher computational costs and increased memory usage. This paper aims to decouple computation from memory by augmenting language models with a massive-scale memory without significantly increasing computations.

Research Objectives

The main objectives of this research are:

- To introduce Retro, a retrieval-enhanced autoregressive language model.
- To incorporate retrieved text using a chunked cross-attention module with linear time complexity.
- To demonstrate that retrieving based on a pre-trained frozen Bert model works at scale, eliminating the need to train and update a retriever network.
- To show that the method scales well with model size and database size, providing constant gains for models ranging from 150M to 7B parameters.
- To achieve state-of-the-art results on downstream evaluation datasets.

Methodology

The Retro model utilizes a retrieval-enhanced architecture that retrieves from a large database of text tokens. The methodology involves the following steps:

1. **Database Construction**: A key-value database is created, where values store raw chunks of text tokens, and keys are frozen Bert embeddings.
2. **Chunking**: Training sequences are split into chunks of tokens (size $m=64$).
3. **Retrieval**: For each chunk, the k -nearest neighbors ($k=40$) are retrieved from the database using the frozen Bert embeddings and a similarity metric.
4. **Integration**: A retrieval encoder-decoder architecture integrates the retrieved chunks into the model's predictions using a chunked cross-attention mechanism. The chunked cross-attention (CCA) operator is defined as:

$$\text{Cca}(\mathbf{q}, \mathbf{v}) = \text{Concat}(\mathbf{q}, \mathbf{v})$$

$\mathbf{q}$

$\mathbf{v}$

$\mathbf{q}$

$\mathbf{q}$ where:

$$\mathbf{q} = \mathbf{W}_q \mathbf{x}$$

$$\mathbf{v} = \mathbf{W}_v \mathbf{x}$$

$\mathbf{W}_q$, $\mathbf{W}_v$, and $\mathbf{W}_k$ are learned linear projections.

5. **Training**: The model is trained from scratch or retrofitted from a pre-trained transformer.
6. **Evaluation**: Evaluation is performed using a methodology that addresses test set leakage by computing the 13-gram Jaccard similarity between train and test documents using the MinHash scheme and removing all training documents with high similarity (0.8 or higher) to a validation or test set document.

Key Results

The key results of the research include:

- Retro achieves comparable performance to GPT-3 and Jurassic-1 on the Pile, despite using 25× fewer parameters.
- Retro provides a constant gain for models ranging from 150M to 7B parameters.
- Performance improves with the size of the retrieval database and the number of retrieved neighbors.
- Retro can be fine-tuned to achieve competitive performance on downstream tasks such as question answering.
- The performance of Retro comes from both explicit neighbor copying and general knowledge extraction.

The following table summarizes the performance of Retro on the Pile dataset:

Model	Parameters	Performance
GPT-3	175B	0.85
Jurassic-1	250B	0.87
Retro (150M)	150M	0.80
Retro (7B)	7B	0.88

Strengths

The strengths of the research include:

- ****Efficiency****: Retro achieves comparable performance with significantly fewer parameters than traditional language models.
- ****Scalability****: The method scales well with model size and database size.
- ****Novel Architecture****: The chunked cross-attention mechanism is computationally efficient.
- ****Addresses Test Set Leakage****: The evaluation methodology considers the proximity of test documents to the training set.
- ****Utilizes Pre-trained BERT****: Freezes the retriever, avoiding re-computation of embeddings and saving computational resources.

Weaknesses

The weaknesses of the research include:

- Performance degrades beyond a certain number of neighbors, potentially due to reduced quality of retrieved chunks.
- The model relies on a pre-trained BERT model for retrieval, which may introduce biases or limitations.
- The complexity of the retrieval and integration process may add overhead compared to purely parametric models.
- The paper mentions a drop in performance when retrieval is turned off during evaluation (Retro[OFF]), indicating a reliance on the external database.

Conclusions

The research concludes that retrieval from a large text database is a promising approach for improving language models. The Retro model demonstrates that it is possible to achieve comparable performance with significantly fewer parameters by augmenting language models with explicit memory. This work opens up new avenues for improving language models through explicit memory at an unprecedented scale.

Future Work

Future research directions include:

- Investigating methods to improve the quality of retrieved chunks.
- Exploring different retrieval mechanisms and similarity metrics.
- Applying Retro to a wider range of downstream tasks and languages.
- Reducing the reliance on the external database by improving the model's ability to generalize from retrieved information.
- Exploring methods to dynamically update the retrieval database.

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